

Robson Christie

Curriculum Vitae



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PROFILE

Researcher in theoretical and computational physics developing open quantum systems approaches to problems in cosmology, quantum gravity, and astrophysics. My work combines Lindblad dynamics, stochastic unravellings, decoherence theory, quantum field theory, and numerical simulation to understand how environmental interactions shape the behaviour of quantum fields in gravitational settings. I am particularly interested in using open-system methods to connect microscopic quantum dynamics with observable signatures in the early universe and high-energy astrophysical environments, and in building reproducible computational tools that turn these questions into testable calculations.

EMPLOYMENT HISTORY

MARCH 2024 – MARCH 2026 (FT)

University of Portsmouth

Research Fellow

Research on open quantum systems across quantum simulation and early-universe cosmology. Developed circuit-level protocols and measurement strategies for simulating dissipative dynamics on IBM quantum hardware, with benchmarking against classical open-system solvers. Built and ran lattice simulations of quantum fields during inflation to study coarse-graining, non-adiabatic excitation, and decoherence-induced stabilisation of metastable states. Teaching contributions included weekly Maths Helpdesk support and delivering Quantum Mechanics lectures.

MARCH 2023 – MARCH 2024 (FT)

Imperial College London

EPSRC Research Associate in Mathematics

Supported grant EP/W522673/1 by developing theory and algorithms for quantum rare-event simulation in stochastic trajectory formulations. Derived practical estimators for transition rates and path probabilities, connected open-system dynamics to large-deviation ideas, and validated methods on representative models. Delivered maintainable research software and analysis tooling, and collaborated across mathematics/physics/chemical-physics communities to translate theory into simulation methods.

SEPT 2018 – MARCH 2023 (FT)

Imperial College London

Graduate student (PhD awarded March 2023)

Thesis: *Gaussian dynamics in the Lindblad equation and its stochastic unravellings* (Advisor: Dr Eva Maria Graefe). Doctoral research developed analytical and numerical tools connecting quantum open-system dynamics to classical stochastic descriptions. Key contributions included: closed-form evolution for Gaussian states under quadratic Hamiltonians and linear Lindbladians; systematic comparisons of continuous versus jump unravellings at the trajectory level; and semiclassical limits for broad classes of open quantum systems. Built simulation codes and analysis workflows (Matlab/Python/Mathematica) to validate theory and communicate results through publications and talks.

PROGRAMMING SKILLS

SCIENTIFIC COMPUTING	Python (advanced), Matlab (advanced), Mathematica (advanced), C++ (novice), , CUDA (novice)
QUANTUM COMPUTING	IBM Qiskit (advanced), QuTiP (advanced)
HPC & WORKFLOW	Git/GitHub, Slurm, PBS
SCIENTIFIC TOOLS	L ^A T _E X, ORCA, TeraChem, Quartus, LTSpice

EDUCATION

- **PhD** (Mathematics) — Imperial College London
Advisor: Dr Eva Maria Graefe
Sept 2018 – March 2023
- **MSc** (Quantum Fields and Fundamental Forces) — Imperial College London
Advisor: Prof Jonathan Halliwell
Sept 2017 – Sept 2018
- **BMat&Phys** (First class honours) — The University of Manchester
Sept 2014 – July 2017
- **A Levels** — Maths A*, Physics A*, Chemistry A*, Further Maths A, Biology A
2012 – 2014

CURRENT PERSONAL PROJECT

Minkowski Kart: Relativistic Racing

Personal project

github.com/rchristie95/MinkowskiKart

Building a relativistic racing game by forking SuperTuxKart and integrating ideas from OpenRelativity to simulate visual and gameplay effects inspired by special and general relativity. Implemented custom GLSL shader effects and C++ gameplay logic for nine relativistic power-ups, including black holes, time-dilation, cosmic strings, and Lorentz-grid frame shifts. This is a fun way for me to learn C++ and the first working version is available to download from my github: [MinkowskiKart](#).

Christie, R., Joo, J., Kaplanek, G., Vennin, V., Wands, D. (in preparation). Quantum Stochastic Inflation, *Manuscript in preparation*

Summary: Formulates stochastic inflation in an open-quantum-system framework, deriving a GKLS dynamics for coarse-grained scalar field variables and showing how Hubble friction and diffusion arise from the same quantum channel while recovering classical stochastic-inflation Fokker–Planck dynamics in the light-field regime.

Christie, R., Murray, C., Kim, Y., Joo, J. (2026). Non-Equilibrium Dynamics of the Time-Dependent Excitonic Coupling in Fluorescent Protein Dimers, *arXiv preprint*, [arXiv:2605.00027](#)

Summary: Combined transition-density coupling calculations with stochastic quantum simulations to quantify strong near-field excitonic coupling in dimeric Venus fluorescent protein and model the post-absorption transition from delocalised excitation to incoherent hopping.

Christie, R., Joo, J., Kaplanek, G., Vennin, V., Wands, D. (2025). Cosmic Lockdown: When Decoherence Saves the Universe from Tunneling, *arXiv preprint*, [arXiv:2512.14204](#)

Summary: Lattice simulations of a coarse-grained inflationary field show how non-adiabatic dynamics can populate false vacua; decoherence sourced by heavy modes then suppresses subsequent tunnelling, dynamically stabilising metastable states.

Christie, R., Limmer, D.T., Bolhuis, P.G. (2025). Transition Path and Interface Sampling of Stochastic Schrödinger Dynamics, *The Journal of Chemical Physics*, vol. 162, no. 11, p. 114120, 2025; [arXiv:2411.00490](#)

Summary: Extended path & interface sampling to quantum trajectories, enabling efficient estimation of transition rates and revealing non-Arrhenius behaviour at low temperature linked to measurement backaction (anti-Zeno effects).

Christie, R., Baek, K., Bang, J., Joo, J. (2024). Quantum Simulation of Dynamical Transition Rates in Open Quantum Systems, *arXiv preprint*, [arXiv:2412.17229](#)

Summary: Developed a quantum-simulation protocol to estimate open-system transition rates without full Lindblad evolution by decomposing rates into independently measurable contributions; validated on IBM quantum hardware.

Christie, R., Trayford, J. (2024). The Sound of Decoherence, **Sonification Award Winner 2025**, [arXiv:2412.17045](#); [sonification.design/Decoherence](#)

Summary: Created an audio mapping from density-matrix structure to perceptual features (energy levels as tones, coherences as binaural cues), enabling listeners to track tunnelling, thermalisation, decoherence, and recoherence in Lindblad dynamics.

Christie, R., Eastman, J. (2024). Quantum Tunnelling and Thermally Driven Transitions in a Double Well Potential at Finite Temperature, *Journal of Physics A: Mathematical and Theoretical*, vol. 57, no. 23, p. 235005, 2024; [arXiv:2312.12101](#)

Summary: Mapped the thermal-to-quantum tunnelling crossover in a dissipative double well, showing how low-temperature noise can randomise yet preserve tunnelling, and introduced a Lindblad Brownian-motion model that removes the zero-temperature singularity.

Christie, R., Eastman, J., Schubert, R., Graefe, E.M. (2022). Quantum-jump vs stochastic Schrödinger dynamics for Gaussian states with quadratic Hamiltonians and linear Lindbladians. *Journal of Physics A: Mathematical and Theoretical*, vol. 55, no. 45, p. 455302, 2022; [arXiv:2203.11530](#)

Summary: Compared continuous (stochastic Schrödinger) and jump unravellings of Lindblad dynamics, showing identical ensemble limits but qualitatively different single-trajectory behaviour. Derived closed-form Gaussian evolution for continuous trajectories and introduced a Gaussian Hagedorn-wavepacket scheme for jump trajectories, validated on measured and damped oscillators.